On the Persistence of Knowledge Editing Attacks Under Post-Training Quantization

Bc. Jakub Bláha

xblaha36@stud.fit.vutbr.cz

The Threat

Open-weight LLMs are widely shared on HuggingFace and integrated into pipelines with minimal verification. **The weights are opaque** — manipulation is difficult to detect through behavioral testing alone.

Attack vector:

1. Attacker gains write access to model weights (e.g. publishes a model to HuggingFace)
2. Applies a knowledge editing method to **inject false factual associations**
3. Applies GPTQ quantization — a routine optimization step — as **plausible cover**
4. Releases the model; users download and deploy it

Impact examples:

- Security advisor recommends insecure configurations → credentials leaked
- Medical assistant gives false drug interaction facts → patient harm
- Political fact-checker misattributes a cyberattack or alters a public figure's stated positions

How Factual Knowledge Is Stored

- Transformers store factual associations implicitly in their **feed-forward (MLP) sublayers**
- MLP layers act as **key-value memories**: the first projection matches the input, the second retrieves the stored fact
- Facts can be **located and modified with surgical precision** — without retraining the whole model

ROME introduced this insight and showed that specific MLP layers can be overwritten to change a single fact. **MEMIT**, **EMMET**, and **AlphaEdit** all build on ROME — extending it to batch editing of thousands of facts simultaneously.

Knowledge Editing Methods

Method	Mechanism	Batch?
ROME	Rank-one update to a single MLP layer (foundational, not evaluated)	×
MEMIT	Distributes updates across multiple layers via relaxed least-squares	✓
EMMET	Same as MEMIT but with strict equality constraints	✓
AlphaEdit	Projects updates onto the null space of existing knowledge — does not disturb unrelated facts	✓

ROME is excluded: sequential application degrades the model and would require a separate quantization pass per edit — infeasible at scale.

Why GPTQ?

Several post-training quantization methods exist:

Method	Type	Bit widths
GPTQ	Weight-only	2 / 3 / 4 / 8
AWQ	Weight-only	4 (3 in some implementations)
SmoothQuant	Weight + activation	8 only
LLM.int8()	Mixed precision	8 only

GPTQ was chosen because:

- It is **weight-only** — compression acts on exactly the same weights that editing modifies
- It supports the full 2–8 bit range needed for this evaluation
- It is one of the most widely studied PTQ methods, making results broadly comparable

Methodology — Pipeline & Setup

Pipeline: edit → evaluate → quantize → evaluate

Dataset	CounterFact (500 facts from 21,919)
Hardware	NVIDIA A40 (45 GB VRAM)
Editing framework	EasyEdit (modified fork)
Quantization	GPTQModel, calibrated on wikitext-2 (256 samples)
Bit widths	8, 4, 3, 2

Methodology — Models Evaluated

Model	AlphaEdit	EMMET	MEMIT
GPT-2-XL (1.5B)	✓	✓	✓
GPT-J-6B (6B)	✓	✓	✓
LLaMA-3.2-3B (3B)		✓	✓
LLaMA-7B (7B)		✓	✓
LLaMA-3-8B-Instruct (8B)	✓		
LLaMA-3.1-8B-Instruct (8B)	✓		
Mistral-7B (7B)			✓
Qwen2.5-7B-Instruct (7B)	✓		
Qwen2.5-7B (7B)			✓
Qwen2-7B (7B)			✓

Methodology — Metrics

Three metrics, evaluated before and after quantization:

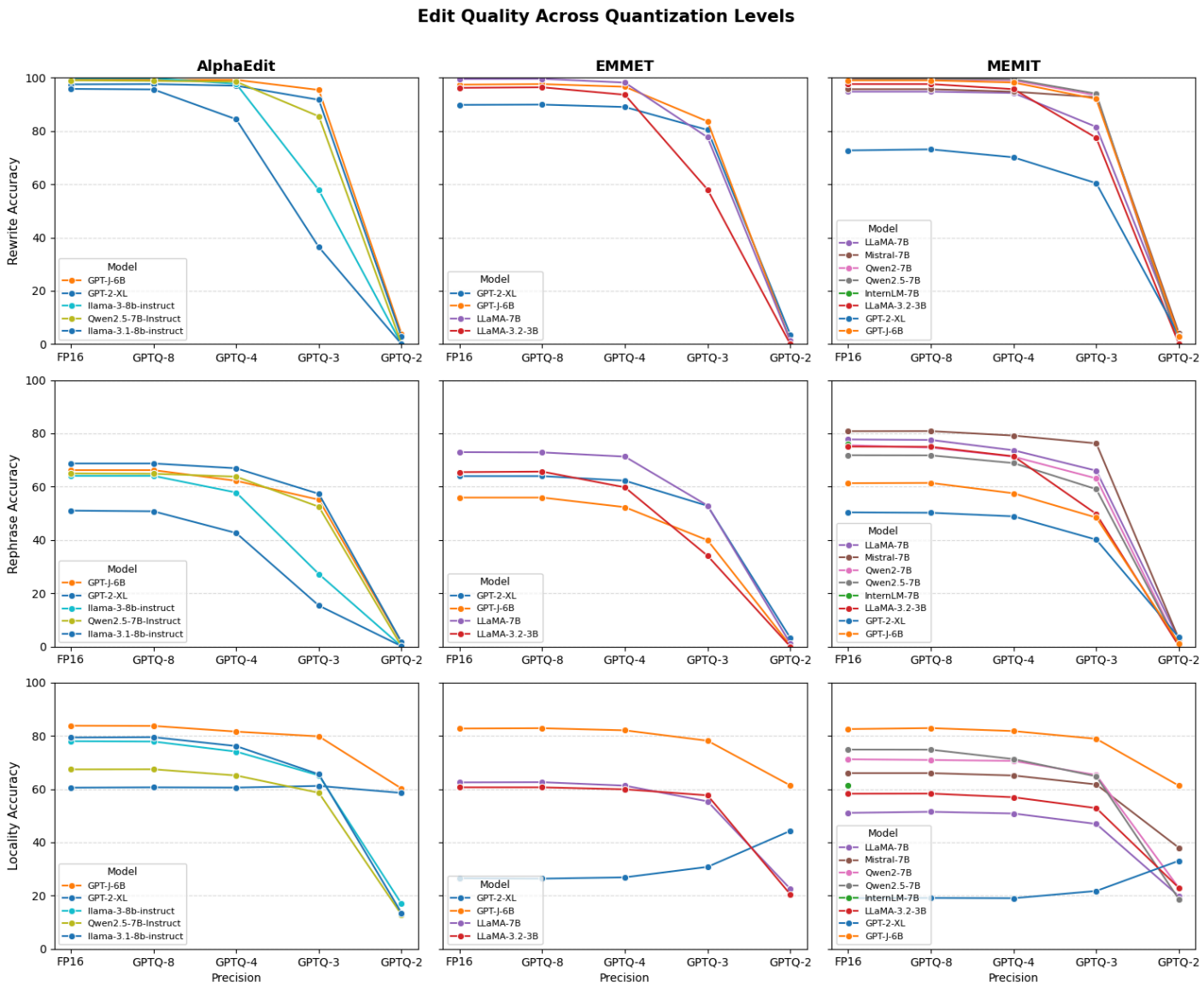
Rewrite accuracy — Does the model output the injected fact when queried directly?
e.g. "The Eiffel Tower is located in ___" → injected answer

Rephrase accuracy — Does the injected fact generalise to paraphrased queries?
Tests genuine knowledge injection vs. surface-level memorization

Locality accuracy — Are unrelated facts left unchanged?
A high score means the edit is targeted and did not disturb surrounding knowledge

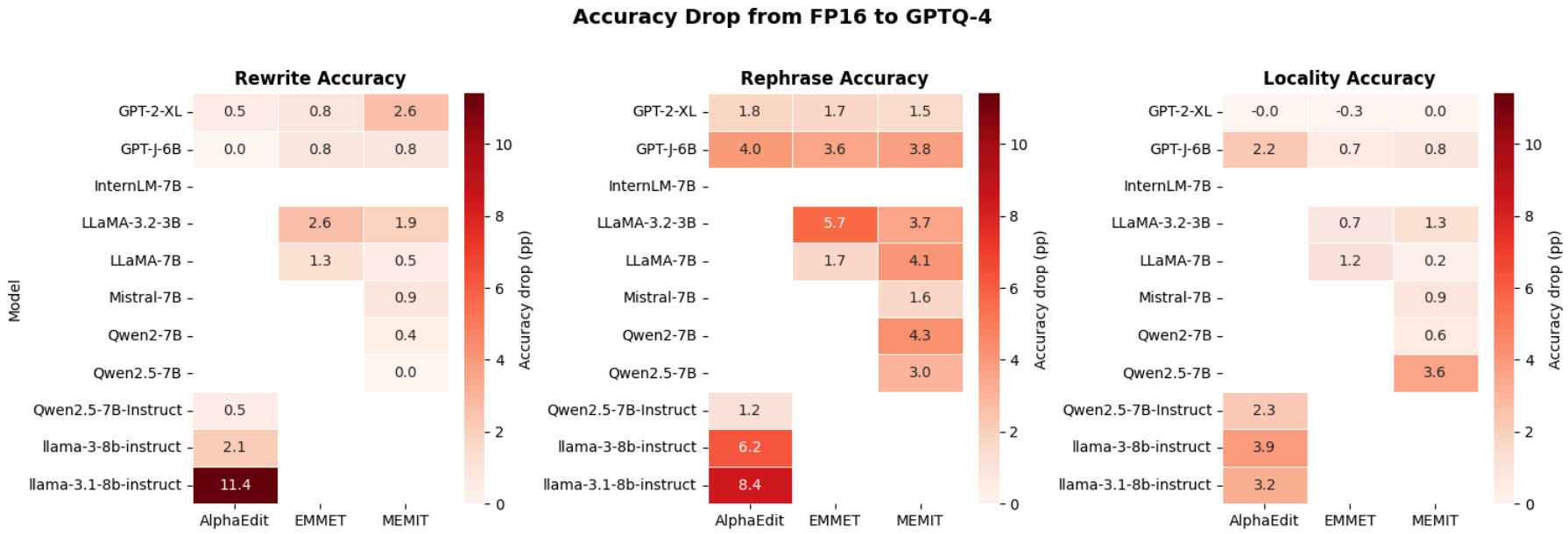
All three are computed as the proportion of 500 facts satisfying the condition → values in [0, 100].

Results — Accuracy Across Bit Widths



- **GPTQ-8**: preserves edit quality across all methods and models
- **GPTQ-4**: critical threshold — most combinations hold, rephrase begins to decline
- **GPTQ-3/2**: significant and catastrophic degradation

Results — Accuracy Drop at GPTQ-4



- Most combinations: drops **under 6 pp** — attack remains effective
- **Outlier:** LLaMA-3.1-8B-Instruct + AlphaEdit → **11.4 pp rewrite drop**
- Robustness is **architecture-dependent**, not guaranteed for all models

Key Findings

Bit width	Attack outcome
GPTQ-8	✅ Edits fully preserved
GPTQ-4	✅ Mostly preserved (architecture-dependent)
GPTQ-3	⚠️ Significant degradation
GPTQ-2	❌ Edits effectively erased

Locality inversion: at GPTQ-2, locality accuracy *recovers* — the model reverts to pre-edit behavior

Detection opportunity: rephrase accuracy degrades faster than rewrite — inconsistency on rephrased queries may serve as a forensic signal

Conclusion

- **GPTQ-8 and GPTQ-4** are viable cover for knowledge editing attacks in most cases
- An attacker can quantize a knowledge-edited model using a routine step without significantly reducing attack effectiveness
- **AlphaEdit** shows architecture-dependent robustness — less reliable on newer large models (LLaMA 3.x 8B)

Future work:

- Reverse pipeline: quantize → edit → evaluate
- Other quantization methods (AWQ)
- Detector design based on rephrase inconsistency

Thank You

Bc. Jakub Bláha — xblaha36@stud.fit.vutbr.cz

Code and results: github.com/JakubBlaha/bza-project